

Project Title: *Mission Maru Ulavu*

Company: *NirUlavu*

Sector: AgriTech | Robotics | Rural Employment | Sustainable Farming

"Maru Ulavu – Not Just Automation, It's Upliftment"

Overview:

Agriculture in Tamil Nadu is facing a critical shortage of manual labor, particularly during peak seasons for ploughing, sowing, weeding, and harvesting. Younger generations are increasingly moving away from farming, while older farmers struggle to complete essential tasks on time. This results in reduced productivity, increased costs, and, in some cases, abandonment of farmlands.

Mission Maru Ulavu presents a practical, ethical solution: deploying **multi-tasking agricultural robots** on a **rental/service basis**, supported by a team of trained local engineers and technicians. This innovation aims to **assist farmers without replacing existing labor**, and crucially, it creates new skilled job opportunities for unemployed engineers in rural areas.

Key Features:

- **Robot-as-a-Service model:** Farmers rent robots as needed, avoiding the high cost of ownership
 - **Multi-functional robots:** Capable of ploughing, sowing, weeding, and spraying
 - **Engineer-backed support:** Trained local engineers maintain and operate the robots, ensuring minimal downtime
 - **Job preservation and creation:** Existing agricultural workers are trained as robot operators, and unemployed skilled engineers find meaningful rural employment
 - **Scalable and customizable:** Suitable for small, medium, and large-scale farms
-

Objectives:

- Improve farming efficiency and reduce delays in critical agricultural operations
- Provide affordable and accessible robotic support to farmers in Tamil Nadu
- Generate new employment avenues for skilled engineers and local laborers
- Promote sustainable and modern agricultural practices while respecting traditional farming communities
- Guarantee **zero job loss** among current agricultural laborers

Social and Economic Impact:

- Accelerates farming tasks by 25–40%, improving overall productivity
- Reduces operational costs by up to 30% for farmers through efficient labor support
- Creates 2–3 new jobs per robot deployed, including skilled engineering roles
- Encourages youth engagement in agriculture through technology-based employment
- Demonstrates that innovation can uplift rural communities by combining technology with human skills

Core Principle:

“Technology should enhance livelihoods, not replace them.”

Mission Maru Ulavu, powered by **NirUlavu**, is a socially responsible agri-robotics initiative that supports Tamil Nadu’s farmers with cutting-edge technology, while simultaneously creating new job opportunities for unemployed skilled engineers and preserving the dignity and employment of traditional agricultural laborers.

● First Step in Mission Maru Ulavu: Solving Nattru Naduthal Problem

"We don't replace people. We support them with machines and create new jobs too."

The Problem – Nattru Naduthal (Paddy Transplanting)

- Nattru Naduthal means planting young paddy saplings into wet fields.
 - It needs **many workers** at the **same time** and must be done **within a few days**.
 - **Today's issues:**
 - **Not enough laborers** – young people are leaving villages.
 - **High cost** – wages per worker are ₹500–₹700 daily.
 - **Time pressure** – delay causes lower crop yield.
 - **Physical stress** – bending in muddy water is hard work.
 - **Uncertainty** – farmers don't know if enough workers will come on time.
-

The Solution – "NirNadu" (நிர்நடு) Robot for Nattru Naduthal

We will build a special **robotic machine** only for doing paddy transplanting, to help farmers in this critical step.

How it works:

- Farmer **rents the robot** for a few days during planting season.
 - A **trained local engineer** comes with the robot to operate it.
 - The robot plants the saplings **quickly and neatly**, just like human workers.
 - One robot can **do the work of 10–15 people** per day.
-

Benefits to Farmers

- **Saves Money** – fewer workers needed.
 - **Saves Time** – faster than humans.
 - **No need to own the robot** – just rent and use.
 - **Reliable** – robot comes with technical support.
-

First Step Toward a Bigger Vision

This robot will be the **first product** under the **Mission Maru Ulavu** plan. Step by step, we will make more machines for other farm tasks like:

- Ploughing
- Weeding
- Harvesting

But we start small — with one focused solution — and grow from there.
